

BELO HORIZONTE CONFINS, Brazil

WMO: 835870

Lat: 19.634S

Lon: 43.969W

Elev: 2715

StdP: 13.31

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	51.4	53.3	37.5	36.4	73.9	41.0	41.8	71.5	18.4	67.2	16.7	67.1	4.5	180	0.322

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.4	89.8	67.2	87.9	67.7	86.1	67.9	74.5	80.9	73.3	80.4	72.2	79.6	7.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	136.0	77.2	71.6	129.1	76.0	70.1	122.5	74.6	40.0	81.4	38.8	80.6	37.8	79.4	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.0	15.0	13.5	DB	46.4	93.7	2.6	2.6	44.5	95.6	43.0	97.2	41.6	98.6	39.7	100.5	(4)
(5)			WB	43.9	76.5	2.5	3.5	42.1	79.0	40.6	81.1	39.2	83.0	37.4	85.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.8	74.2	74.6	73.5	71.6	67.7	65.7	65.3	67.7	70.7	73.1	72.1	73.2		
	DBStd	4.81	3.46	3.11	2.96	3.31	3.45	3.19	3.27	3.71	4.26	4.43	4.14	3.44		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	110	0	0	0	2	16	29	35	15	7	3	3	0		
	CDD50	7577	751	690	728	647	547	472	475	548	620	716	663	720		
	CDD65	2215	286	270	263	199	98	51	46	98	176	255	217	256		
	CDH74	14415	1873	1857	1524	1100	510	309	349	794	1393	1936	1287	1483		
	CDH80	3651	543	542	344	195	44	13	23	153	414	654	342	384		
(14) Wind	WSAvg	6.4	6.2	6.3	6.1	6.0	5.7	5.6	6.1	6.8	7.5	7.5	6.9	6.4		
(15) Precipitation	PrecAvg	52.9	11.5	6.4	6.0	1.9	1.1	0.3	0.3	0.5	1.7	3.0	8.8	12.0		
	PrecMax	65.0	25.2	12.6	13.9	4.6	3.3	1.6	1.7	1.8	3.6	5.4	15.1	23.1		
	PrecMin	36.0	2.4	2.0	1.1	0.4	0.1	0.0	0.0	0.0	0.0	1.0	4.7	5.9		
	PrecStd	7.6	6.4	2.9	3.4	1.3	0.7	0.4	0.4	0.6	1.1	1.3	2.6	4.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.1	89.9	88.0	86.2	84.0	81.6	82.6	87.6	91.6	93.0	90.0	89.3		
		MCWB	69.8	71.2	70.7	68.6	67.1	63.7	62.1	63.2	64.4	65.0	67.5	69.0		
	2%	DB	87.8	87.8	85.9	84.3	80.9	79.2	80.2	84.0	87.9	89.5	86.3	86.2		
		MCWB	68.9	70.2	70.4	69.0	65.9	63.1	61.4	62.7	63.5	65.5	68.1	69.1		
	5%	DB	85.6	85.9	84.0	82.4	79.0	77.3	77.4	81.0	84.5	86.3	84.1	84.3		
		MCWB	69.3	69.7	70.2	68.8	65.2	62.6	61.3	61.3	63.4	65.8	68.3	69.4		
	10%	DB	82.8	83.9	82.1	80.4	77.0	75.4	75.5	78.7	82.1	83.9	80.9	82.1		
		MCWB	69.0	69.3	69.3	68.4	64.6	62.0	60.5	61.2	63.1	65.8	67.7	69.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.2	75.6	75.7	73.5	70.8	67.8	66.3	67.9	69.8	72.4	73.6	75.0		
		MCDB	81.5	82.1	81.4	80.0	78.4	76.9	75.6	79.2	79.3	81.3	80.3	81.3		
	2%	WB	74.1	73.9	73.6	71.9	69.0	65.9	64.7	65.8	68.1	70.0	71.9	73.2		
		MCDB	80.2	81.7	80.4	79.6	76.9	74.5	74.8	76.6	78.4	79.3	79.0	79.9		
	5%	WB	72.6	72.4	72.1	70.7	67.7	64.7	63.4	64.3	66.6	68.8	70.6	71.9		
		MCDB	79.3	80.9	79.6	78.4	75.7	73.7	73.6	75.6	76.9	78.8	77.8	78.8		
	10%	WB	71.3	71.1	70.9	69.4	66.2	63.4	62.2	63.0	65.5	67.8	69.4	70.7		
		MCDB	78.3	79.6	78.3	76.9	73.3	72.3	72.2	74.2	76.0	78.2	76.5	77.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	17.4	16.5	17.5	18.7	19.9	20.8	21.7	20.9	19.2	16.3	15.9		
		MCDBR	20.0	20.2	19.4	19.6	20.9	22.2	23.6	25.0	25.0	23.4	20.5	19.7		
	5% WB	MCWBR	7.0	6.7	7.0	7.2	8.1	8.5	8.8	8.0	7.6	7.2	6.7	6.9		
		MCWBR	16.4	17.9	17.3	17.7	18.3	19.9	20.7	21.7	19.9	19.1	16.7	16.5		
(40) Clear-Sky Solar Irradiance	taub		0.387	0.377	0.372	0.352	0.334	0.323	0.327	0.357	0.439	0.436	0.399	0.391		
		taud	2.456	2.472	2.509	2.527	2.546	2.576	2.537	2.429	2.197	2.259	2.403	2.444		
	Ebn at Noon		304	303	297	291	285	283	284	285	272	282	298	303		
		Edh at Noon	38	37	35	32	29	27	29	34	46	45	40	39		
(44) All-Sky Solar Radiation	RadAvg		1866	1908	1690	1558	1357	1278	1363	1596	1705	1793	1651	1742		
	RadStd		272	207	150	96	75	80	75	104	168	210	159	182		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	-1.51	-1.33	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	-1.51	-1.33	N/A	N/A	N/A	N/A

Nomenclature: See separate page